

TED(10) 3214

Reg. No.....

REVISION -2015

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/

TECHNOLOGY- OCTOBER 2016

INSTRUMENT TRANSDUCERS

Time:3 Hours

MaxMarks:100

PART- A

(Max Marks:10)

Marks

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. List out any two active transducers
2. Define gauge factor
3. Define Villari effect.
4. Describe photoelectric effect
5. List out two piezo electric materials

(5x2=10)

PART- B

(Max Marks:30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. A strain gauge having a gauge factor of 4 is used for testing a machine. If the gauge resistance is 100Ω , and the strain is 20×10^{-6} , how much will be its new resistance?
2. Derive the sensitivity of linear and rotary potentiometers
3. Describe the working of a RVDT, with a neat sketch.
4. Explain the working of search coil.
5. Explain capacitive level transducer
6. Illustrate the working principle of Photo voltaic cell
7. Explain the functioning of IC Sensor LM35

(5x6=30)

PART- C

(Max Marks:60)

(Answer ONE Full question from each unit. Each question carries 15 marks.)

UNIT- I

- III a) Describe Loading Effect of Potentiometer and its significance in the output. (8)
b) Explain the strain gauge bridge circuit with a neat circuit diagram (7)

OR

- IV a) Derive Gauge factor of a Strain Gauge (8)
b) List out advantages and disadvantages of Semiconductor strain gauges (7)

UNIT- II

- V a) Explain the construction and working of a LVDT (8)
b) Illustrate the working of a Hall Effect transducer (7)

OR

- VI a) Distinguish between Magneto resistive and Magnetostrictive transducers (8)
b) Explain how pressure can be measured using a LVDT (7)

UNIT -III

- VII a) Explain the construction and operational features of a Photo Transistor (8)
b) Explain different applications of Capacitive transducer (7)

OR

- VIII a) Explain the equivalent circuit of a piezo electric transducer (6)
b) Explain Variable Area, Variable Distance and Variable dielectric type capacitive transducers (9)

UNIT- IV

- IX a) Explain the construction and Operation of a Ionisation Chamber (8)
b) Discuss in detail about solid state transducers (7)

OR

- X a) Sketch Schematic diagram of a Geiger Muller Counter and explain its working (8)
b) Summarize features of Nano sensors and its applications in Instrumentation (7)
